

SQL in Metapost

combining technologies

context 2022 meeting

The reason

- So far I had no real reason but there was this question on the list about saving values.
- A natural place to store values is in a Lua table and one can easily save and load these with helpers.
- Less obvious is to use a database, and a complication with for instance most sql databases, is that one has to have a service running.
- A reasonable alternative is SQLite which uses a file and because it is part of the process it has some overhead.
- Of course a database mostly makes sense with large quantities of data that needs to be accessed randomly using different lookup criteria.
- The downside is that one needs to have the library available and but in this case it is a stable one and it does not influence rendering.

Libraries

- When playing with this early in the LuaTeX project we started with native libraries that came with Lua, and in particular MySQL. A simple command line client was used as a start.
- The Lua related libraries were sort of unreliable (and not really maintained) and not implemented that efficiently so I switched to just the main one that came with the client.
- For a while we had SwigLib which works okay but in order to use that an infrastructure is needed and we don't see that as our responsibility so that project sort of faded out.
- In the end in ConTeXt we used a minimal ffi interface in LuaTeX.
- In LuaMetaTeX we have a few so called optional libraries. These are in fact very light-weight interfaces that assume most work to be done in Lua. We don't want dependencies and only these few are officially supported in this way. The SQLite interface is one of them.
- With ConTeXt we ship several Lua support mechanisms for accessing databases. These perform well and we use it for web driven typesetting services (framework).

Prerequisites

We assume that the `sqlite3.dll` is present. Don't forget to remake the file database after putting the file in place. On MS Windows the preferred location is here:

```
1 E:/tex-context/tex/texmf-win64/bin/lib/luametatex/sqlite/sqlite3.dll
```

If at some point loading fails one can turn on trackers:

```
1 \enabletrackers[sql*]
```

```
2 \enabletrackers[*lib*]
```

As with any library, using one that comes from the system can be of any version and who knows what happened to the api.

Walk through the demo: `context-2022-demo-sql-mps.tex`